

BEE 4750 Homework 4: Linear Programming and Capacity Expansion

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Due Date

Thursday, 11/06/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to analytically (or graphically) solve a linear program.
- Problem 2 asks you to formulate and solve a resource allocation problem using linear programming.
- Problem 3 asks you to formulate, solve, and analyze a standard generating capacity expansion problem.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

Activating project at `~/Desktop/BEE 4750/hw4-matthw4`

```
using JuMP
using HiGHS
using DataFrames
using Plots
using Measures
using CSV
using MarkdownTables
```

Problems (Total: 30 Points)

Problem 1 (Total: 3 Points)

Solve the following linear program **analytically**. Show your work and justify any reasoning.

$$\begin{aligned} \max_{x,y} \quad & 3x + 2y \\ \text{subject to:} \quad & x + 4y \leq 16 \\ & x - 2y \leq -1 \\ & 0 \leq x \leq 4 \\ & 1 \leq y \leq 4. \end{aligned}$$

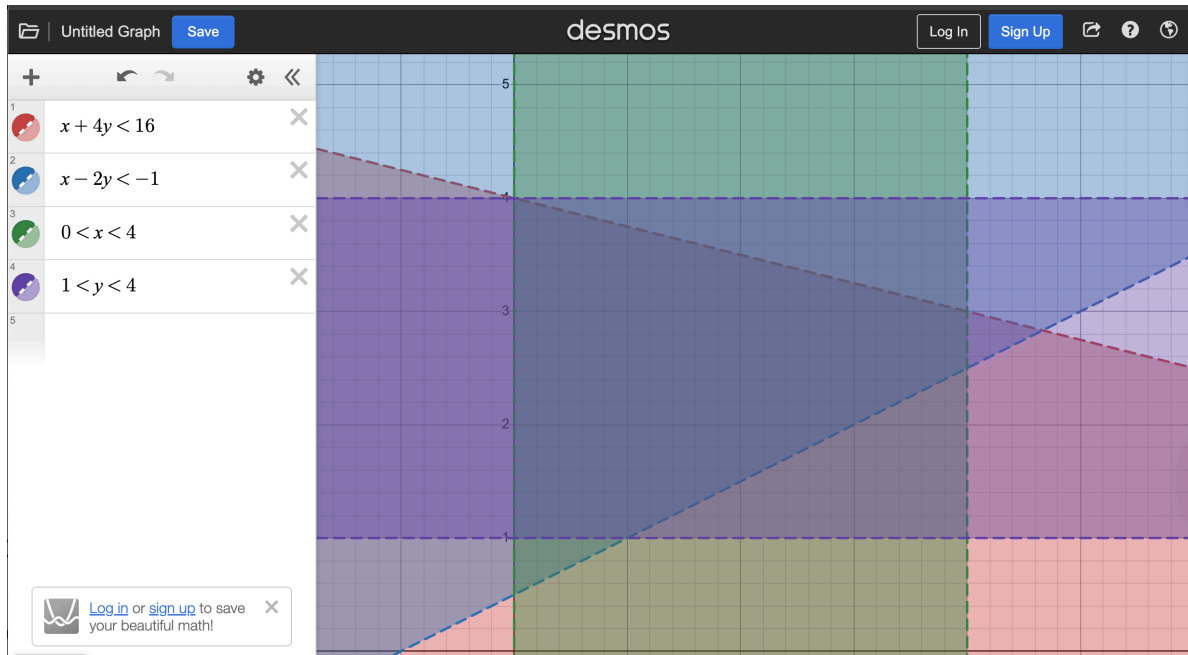


Figure 1: Desmos graph

For linear programs, the optimal solution must lie on an intersection. I graphed the problem in Desmos and tested the objective at all 5 intersections within the feasible space.

Intersection 1: $x=0$ intersects $y=4$ at $(0,4)$

$$3 * 0 + 2 * 4 = 8$$

Intersection 2: $x=0$ intersects $y=1$ at $(0,1)$

$$3 * 0 + 2 * 1 = 2$$

Intersection 3: $x-2y=-1$ intersects $y=1$

$$x - 2(1) = -1$$

$$x = 1$$

$$y = 1$$

$$(1,1)$$

$$3 * 1 + 2 * 1 = 5$$

Intersection 4 $x-2y=-1$ intersects $x=4$

$$4 - 2y = -1$$

$$y = 5/2$$

$$x = 4$$

$$(4, 2.5)$$

$$3 * 4 + 2 * 2.5 = 17$$

Intersection 5: $x+4y=16$ intersects $x=4$

$$4 + 4y = 16$$

$$y = 3$$

$$x = 4$$

$$(4, 3)$$

$$3 * 4 + 2 * 3 = 18$$

Optimal solution at $(4,3)$ with objective function 18

Problem 2 (12 points)

A farmer has access to a pesticide which can be used on corn, soybeans, and wheat fields, and costs \$70/kg-yr to apply. The crop yields the farmer can obtain following crop yields by applying varying rates of pesticides to the field are shown in Table 1.

Application Rate (kg/ha)	Soybean (kg/ha)	Wheat (kg/ha)	Corn (kg/ha)
0	2900	3500	5900
1	3800	4100	6700
2	4400	4200	7900

Table 1: Crop yields from applying varying pesticide rates for Problem 1.

The costs of production, *excluding pesticides*, for each crop, and selling prices, are shown in Table 2.

Crop	Production Cost (\$/ha-yr)	Selling Price (\$/kg)
Soybeans	350	0.36
Wheat	280	0.27

Crop	Production Cost (\$/ha-yr)	Selling Price (\$/kg)
Corn	390	0.22

Table 2: Costs of crop production, excluding pesticides, and selling prices for each crop.

Recently, environmental authorities have declared that farms cannot have an *average* application rate on soybeans, wheat, and corn which exceeds 0.8, 0.7, and 0.6 kg/ha, respectively. The farmer has asked you for advice on how they should plant crops and apply pesticides to maximize profits over 130 total ha while remaining in regulatory compliance if demand for each crop (which is the maximum the market would buy) this year is 250,000 kg?

Problem 2.1

Formulate a linear program for this resource allocation problem, including clear definitions of decision variable(s) (including units), objective function(s), and constraint(s) (make sure to explain functions and constraints with any needed derivations and explanations). **Tip: Make sure that all of your constraints are linear.**

Decision variables:

$$A_{s,0}, A_{s,1}, A_{s,2}, A_{w,0}, A_{w,1}, A_{w,2}, A_{c,0}, A_{c,1}, A_{c,2}$$

$A_{s,0}$ (hectares) is the area of land with soybeans and application rate 0 kg/ha

s = Soybeans, w = Wheat, c = Corn

Objective function: maximize profit

$\max P = \text{selling revenue} - \text{production cost} - \text{pesticide cost}$

selling revenue = $0.36(2900A_{s,0} + 3800A_{s,1} + 4400A_{s,2}) +$

$0.27(3500A_{w,0} + 4100A_{w,1} + 4200A_{w,2}) +$

$0.22(5900A_{c,0} + 6700A_{c,1} + 7900A_{c,2})$

production cost = $350(A_{s,0} + A_{s,1} + A_{s,2}) + 280(A_{w,0} + A_{w,1} + A_{w,2}) + 390(A_{c,0} + A_{c,1} + A_{c,2})$

pesticide cost = $70(0(A_{s,0} + A_{w,0} + A_{c,0}) + 1(A_{s,1} + A_{w,1} + A_{c,1}) + 2(A_{s,2} + A_{w,2} + A_{c,2}))$

Constraints:

Average application rate constraint

$$\frac{0A_{s,0} + 1A_{s,1} + 2A_{s,2}}{A_{s,0} + A_{s,1} + A_{s,2}} \leq 0.8 \rightarrow -0.8A_{s,0} + 0.2A_{s,1} + 1.2A_{s,2} \leq 0$$

$$\frac{0A_{w,0} + 1A_{w,1} + 2A_{w,2}}{A_{w,0} + A_{w,1} + A_{w,2}} \leq 0.7 \rightarrow -0.7A_{w,0} + 0.3A_{w,1} + 1.3A_{w,2} \leq 0$$

$$\frac{0A_{c,0} + 1A_{c,1} + 2A_{c,2}}{A_{c,0} + A_{c,1} + A_{c,2}} \leq 0.6 \rightarrow -0.6A_{c,0} + 0.4A_{c,1} + 1.4A_{c,2} \leq 0$$

Land constraint

$$A_{s,0} + A_{s,1} + A_{s,2} + A_{w,0} + A_{w,1} + A_{w,2} + A_{c,0} + A_{c,1} + A_{c,2} \leq 130$$

Demand constraint

$$2900A_{s,0} + 3800A_{s,1} + 4400A_{s,2} \leq 250,000kg$$

$$3500A_{w,0} + 4100A_{w,1} + 4200A_{w,2} \leq 250,000kg$$

$$5900A_{c,0} + 6700A_{c,1} + 7900A_{c,2} \leq 250,000kg$$

Non-negativity constraint

$$A_{s,0}, A_{s,1}, A_{s,2}, A_{w,0}, A_{w,1}, A_{w,2}, A_{c,0}, A_{c,1}, A_{c,2} \geq 0$$

Problem 2.2

How many ha should the farmer dedicate to each crop and with what pesticide application rate(s)? How much profit will the farmer expect to make?

```
# Create model
model = Model(HiGHS.Optimizer)
set_silent(model)

# Decision variables
@variables(model, begin
    As0 >= 0
    As1 >= 0
    As2 >= 0
    Aw0 >= 0
    Aw1 >= 0
    Aw2 >= 0
    Ac0 >= 0
    Ac1 >= 0
```

```

    Ac2 >= 0
end)

# Objective function components
selling_revenue =
    0.36*(2900*As0 + 3800*As1 + 4400*As2) +
    0.27*(3500*Aw0 + 4100*Aw1 + 4200*Aw2) +
    0.22*(5900*Ac0 + 6700*Ac1 + 7900*Ac2)

production_cost =
    350*(As0 + As1 + As2) +
    280*(Aw0 + Aw1 + Aw2) +
    390*(Ac0 + Ac1 + Ac2)

pesticide_cost = 70*(
    0*(As0 + Aw0 + Ac0) +
    1*(As1 + Aw1 + Ac1) +
    2*(As2 + Aw2 + Ac2))

# Objective: maximize profit
@objective(model, Max, selling_revenue - production_cost - pesticide_cost)

# Constraints
@constraint(model, -0.8*As0 + 0.2*As1 + 1.2*As2 <= 0)
@constraint(model, -0.7*Aw0 + 0.3*Aw1 + 1.3*Aw2 <= 0)
@constraint(model, -0.6*Ac0 + 0.4*Ac1 + 1.4*Ac2 <= 0)

land = @constraint(model, As0 + As1 + As2 + Aw0 + Aw1 + Aw2 + Ac0 + Ac1 + Ac2 <= 130)

@constraint(model, 2900*As0 + 3800*As1 + 4400*As2 <= 250000)
@constraint(model, 3500*Aw0 + 4100*Aw1 + 4200*Aw2 <= 250000)
@constraint(model, 5900*Ac0 + 6700*Ac1 + 7900*Ac2 <= 250000)

# Solve the model
optimize!(model)

# Display results
println("Optimal profit = ", objective_value(model))
println("\nDecision variables:")
for v in all_variables(model)
    println("$ (name(v)) = ", value(v))
end

```

```

end

# ha dedicated to each crop
println("Total soybean area (ha): ", value(As0)+value(As1)+value(As2))
println("Total wheat area (ha): ", value(Aw0)+value(Aw1)+value(Aw2))
println("Total corn area (ha): ", value(Ac0)+value(Ac1)+value(Ac2))

println("Shadow price (dollars): ", shadow_price(land))

```

Optimal profit = 116741.16702082448

Decision variables:

```

As0 = 13.812154696132604
As1 = 55.24861878453038
As2 = 0.0
Aw0 = 6.7433064173395625
Aw1 = 15.734381640458977
Aw2 = 0.0
Ac0 = 26.92307692307692
Ac1 = 0.0
Ac2 = 11.538461538461547
Total soybean area (ha): 69.06077348066299
Total wheat area (ha): 22.477688057798538
Total corn area (ha): 38.46153846153847
Shadow price (dollars): 729.4

```

The problem was solved using JUMP. I inputted a screenshot of my optimization equations into ChatGPT so that I did not have to type the linear program into Julia step by step.

Total soybean area (ha): 69.06077348066299

rate 0 kg/ha: 13.812154696132604 ha

rate 1 kg/ha: 55.24861878453038 ha

rate 2 kg/ha: 0.0 ha

Total wheat area (ha): 22.477688057798538

rate 0 kg/ha: 6.7433064173395625 ha

rate 1 kg/ha: 15.734381640458977 ha

rate 2 kg/ha: 0.0 ha

Total corn area (ha): 38.46153846153847

rate 0 kg/ha: 26.92307692307692 ha
rate 1 kg/ha: 11.538461538461547 ha
rate 2 kg/ha: 0.0 ha
Profit: \$116741.16702082448

Problem 2.3

The farmer has an opportunity to buy an extra 10 ha of land. How much extra profit would this land be worth to the farmer? Discuss why this value makes sense and under what conditions you would recommend the farmer should make the purchase.

```
# Create model
model = Model(HiGHS.Optimizer)
set_silent(model)

# Decision variables
@variables(model, begin
    As0 >= 0
    As1 >= 0
    As2 >= 0
    Aw0 >= 0
    Aw1 >= 0
    Aw2 >= 0
    Ac0 >= 0
    Ac1 >= 0
    Ac2 >= 0
end)

# Objective function components
selling_revenue =
    0.36*(2900*As0 + 3800*As1 + 4400*As2) +
    0.27*(3500*Aw0 + 4100*Aw1 + 4200*Aw2) +
    0.22*(5900*Ac0 + 6700*Ac1 + 7900*Ac2)

production_cost =
    350*(As0 + As1 + As2) +
    280*(Aw0 + Aw1 + Aw2) +
    390*(Ac0 + Ac1 + Ac2)

pesticide_cost = 70*(
    0*(As0 + Aw0 + Ac0) +
```

```

1*(As1 + Aw1 + Ac1) +
2*(As2 + Aw2 + Ac2))

# Objective: maximize profit
@objective(model, Max, selling_revenue - production_cost - pesticide_cost)

# Constraints
@constraint(model, -0.8*As0 + 0.2*As1 + 1.2*As2 <= 0)
@constraint(model, -0.7*Aw0 + 0.3*Aw1 + 1.3*Aw2 <= 0)
@constraint(model, -0.6*Ac0 + 0.4*Ac1 + 1.4*Ac2 <= 0)

@constraint(model, As0 + As1 + As2 + Aw0 + Aw1 + Aw2 + Ac0 + Ac1 + Ac2 <= 140)

@constraint(model, 2900*As0 + 3800*As1 + 4400*As2 <= 250000)
@constraint(model, 3500*Aw0 + 4100*Aw1 + 4200*Aw2 <= 250000)
@constraint(model, 5900*Ac0 + 6700*Ac1 + 7900*Ac2 <= 250000)

# Solve the model
optimize!(model)

# Display results
println("Optimal profit = ", objective_value(model))
println("\nDecision variables:")
for v in all_variables(model)
    println("$ (name(v)) = ", value(v))
end

# ha dedicated to each crop
println("Total soybean area (ha): ", value(As0)+value(As1)+value(As2))
println("Total wheat area (ha): ", value(Aw0)+value(Aw1)+value(Aw2))
println("Total corn area (ha): ", value(Ac0)+value(Ac1)+value(Ac2))

print(124035.16702082449-116741.16702082448)

```

Optimal profit = 124035.16702082449

Decision variables:

As0 = 13.8121546961326

As1 = 55.24861878453038

As2 = 0.0

Aw0 = 9.743306417339564

```

Aw1 = 22.734381640458977
Aw2 = 0.0
Ac0 = 26.923076923076938
Ac1 = 0.0
Ac2 = 11.538461538461544
Total soybean area (ha): 69.06077348066297
Total wheat area (ha): 32.477688057798545
Total corn area (ha): 38.46153846153848
7294.0000000000015

```

I solved this by changing 130 ha to 140 ha in the land constraint. The new profit is \$124035.16702082449, which is \$7294.00 extra profit per year. This makes sense because it is 10 times the shadow price of the linear program in problem 2.2. It also makes sense because land is a binding constraint, and increase the amount of land should increase the profit. I would recommend that the farmer make the purchase if the cost of land will be less than the profit. If the cost of land is more than \$7294/yr, then it doesn't make sense to make the purchase. If the cost of land is less than \$7294/yr, then it does make sense.

Problem 3 (15 points)

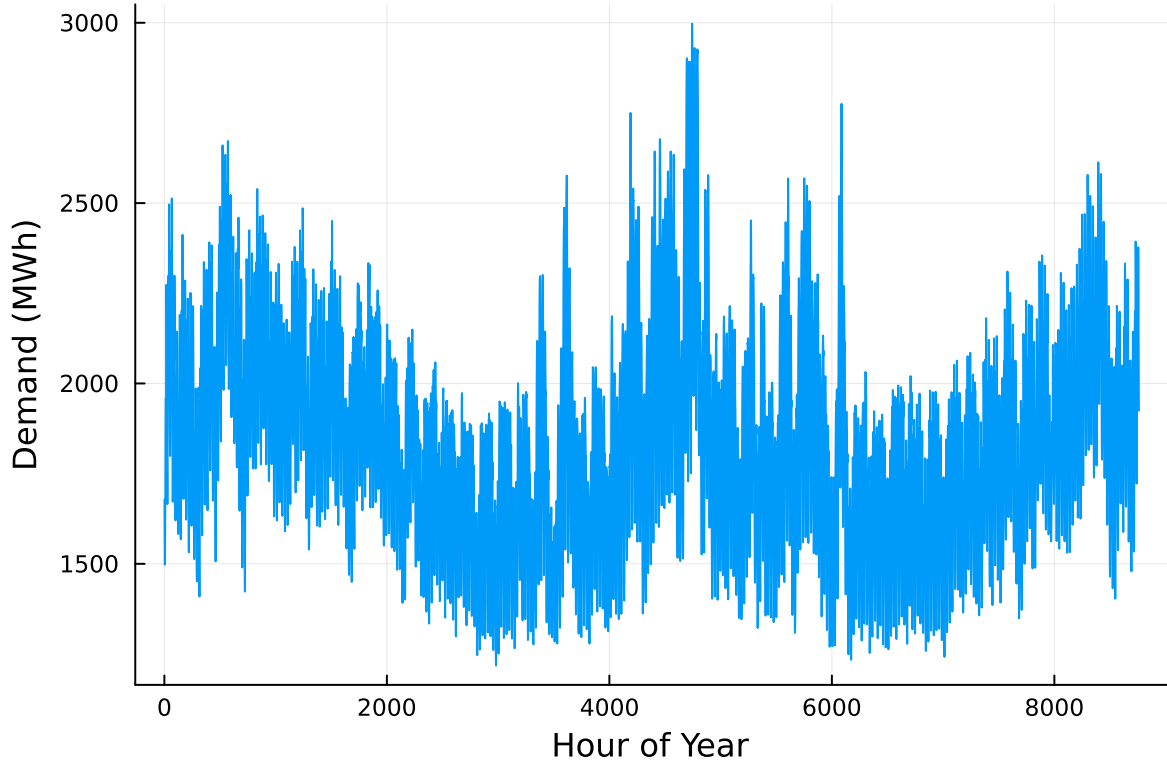
For this problem, we will use hourly load (demand) data from 2013 in New York's Zone C (which includes Ithaca). The load data is loaded and plotted below in Figure 1.

```

# load the data, pull Zone C, and reformat the DataFrame
NY_demand = DataFrame(CSV.File("data/2013_hourly_load_NY.csv"))
rename!(NY_demand, :Time Stamp => :Date)
demand = NY_demand[:, [:Date, :C]]
rename!(demand, :C => :Demand)
demand[:, :Hour] = 1:nrow(demand)

# plot demand
plot(demand.Hour, demand.Demand, xlabel="Hour of Year", ylabel="Demand (MWh)", label=:false)

```



Next, we load the generator data, shown in Table 3. This data includes fixed costs (\$/MW installed), variable costs (\$/MWh generated), and CO2 emissions intensity (tCO2/MWh generated).

```
gens = DataFrame(CSV.File("data/generators.csv"))
```

	Plant	FixedCost	VarCost	Emissions
	String15	Int64	Int64	Float64
1	Geothermal	450000	0	0.0
2	Coal	220000	24	1.0
3	NG CCGT	82000	30	0.43
4	NG CT	65000	40	0.55
5	Wind	91000	0	0.0
6	Solar	70000	0	0.0

Finally, we load the hourly solar and wind capacity factors, which are plotted in Figure 2. These tell us the fraction of installed capacity which is expected to be available in a given hour for generation (typically based on the average meteorology).

```

# load capacity factors into a DataFrame
cap_factor = DataFrame(CSV.File("data/wind_solar_capacity_factors.csv"))

# plot January capacity factors
p1 = plot(cap_factor.Wind[1:(24*31)], label="Wind")
plot!(cap_factor.Solar[1:(24*31)], label="Solar")
axis!("Hour of the Month")
axis!("Capacity Factor")

p2 = plot(cap_factor.Wind[4344:4344+(24*31)], label="Wind")
plot!(cap_factor.Solar[4344:4344+(24*31)], label="Solar")
axis!("Hour of the Month")
axis!("Capacity Factor")

display(p1)
display(p2)

```

You have been asked to develop a generating capacity expansion plan for the utility in Riley County, NY, which currently has no existing electrical generation infrastructure. The utility can build any of the following plant types: geothermal, coal, natural gas combined cycle gas turbine (CCGT), natural gas combustion turbine (CT), solar, and wind. The NY state legislature is planning to enact an annual CO₂ limit, which for the utility would limit the emissions in its footprint to 1.5 MtCO₂/yr.

While coal, CCGT, and CT plants can generate at their full installed capacity, geothermal plants operate at maximum 85% capacity, and solar and wind available capacities vary by the hour depend on the expected meteorology. The utility will also penalize any non-served demand at a rate of \$10,000/MWh.

Problem 3.1

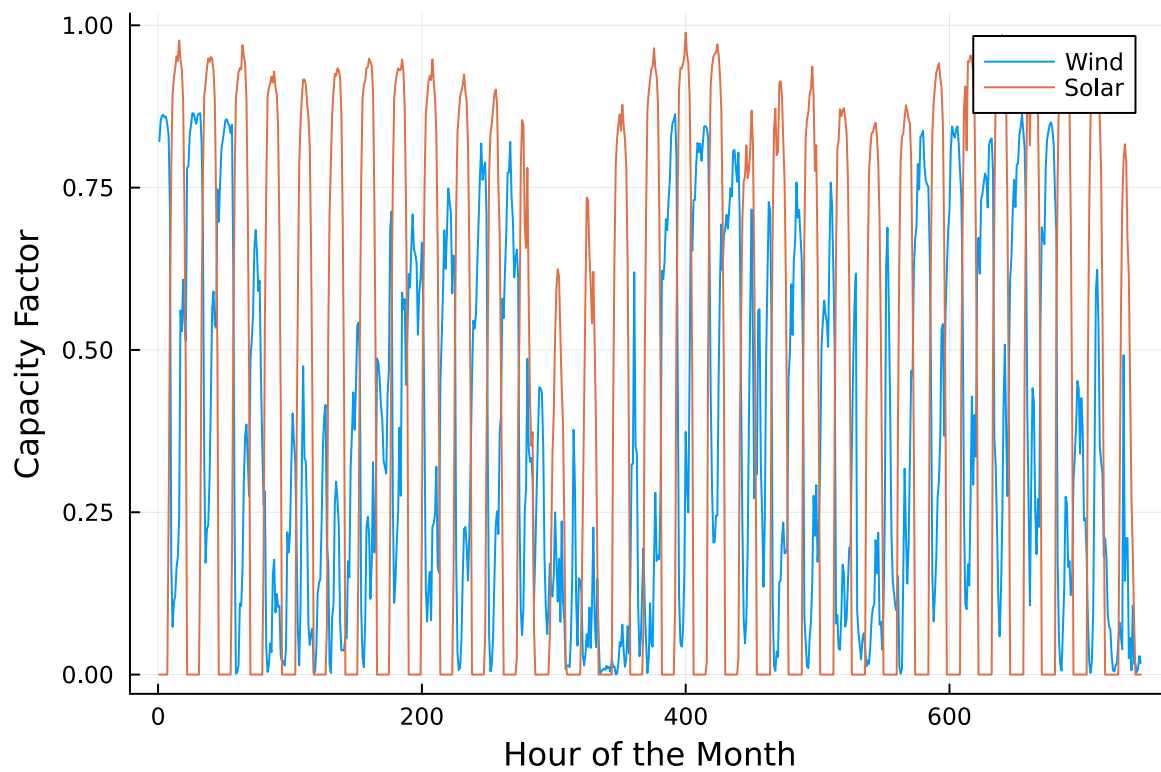
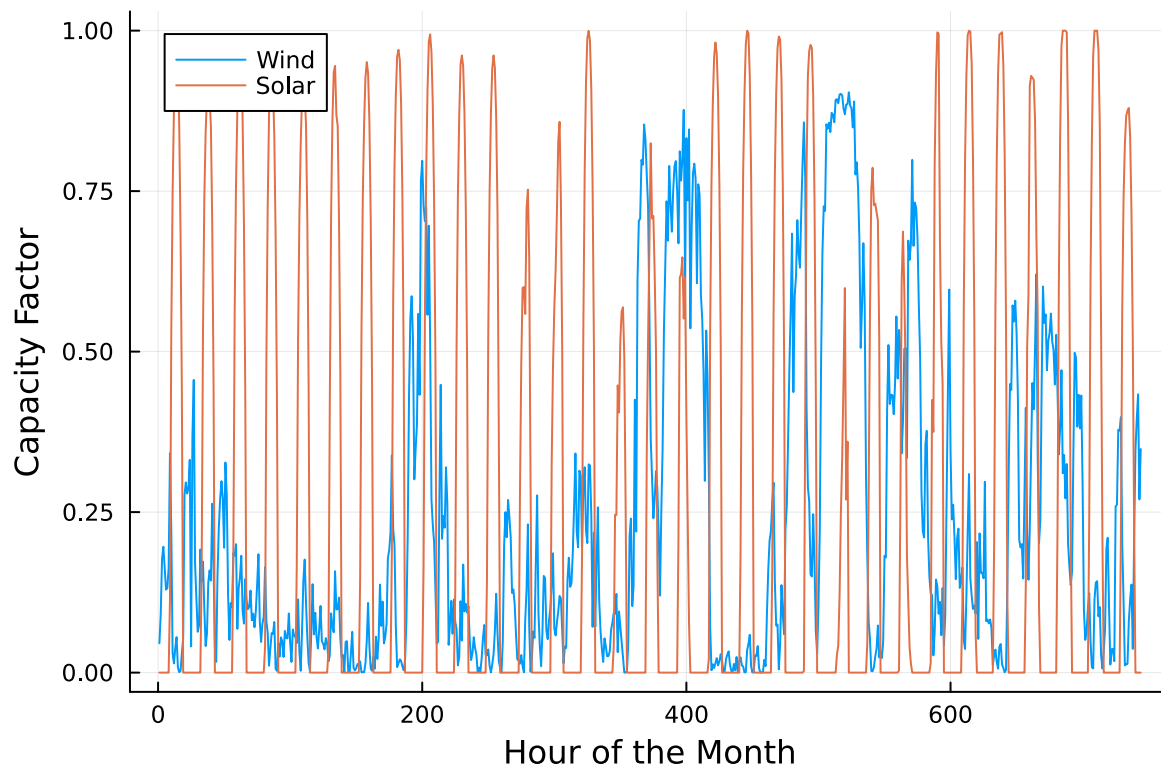
Formulate a linear program for this capacity expansion problem, including clear definitions of decision variable(s) (including units), objective function(s), and constraint(s) (make sure to explain functions and constraints with any needed derivations and explanations).

Geothermal, Coal, NG CCGT, NG CT, Wind, Solar are 1-6 respectively

Decision variables:

$x_1, x_2, x_3, x_4, x_5, x_6$

x_g is the installed capacity of plant type g (MW)



$$y_{g,t} \rightarrow \forall g \in G, \forall t \in T$$

$y_{g,t}$ is the production from plant type g at time t (MWh)

$$NSE_t \rightarrow \forall t \in T$$

NSE_t is the non-served energy (MWh) in hour t

Objective function:

$$\min_{x,y,NSE} \text{TotalCost} = \sum_{g \in G} \text{FixedCost}_g x_g + \sum_{t \in T} \sum_{g \in G} \text{VarCost}_g y_{g,t} + \sum_{t \in T} \text{NSECost} NSE_t$$

Constraints:

$$\text{Demand constraint: } \sum_{g \in G} y_{g,t} + NSE_t \geq d_t, \forall t \in T$$

$$\text{Capacity constraint: } y_{g,t} \leq x_g * c_{g,t}, \forall g \in G, \forall t \in T$$

where $c_{g,t}$ is the capacity factor in time period t for generator type g

$$\text{Non-negativity constraint: } x_g, y_{g,t}, NSE_t \geq 0, \forall g \in G, \forall t \in T$$

Problem 3.2

How much should the utility build of each type of generating plant? What will the total cost be? How much energy will be non-served?

```
# define sets
G = 1:nrow(gens)
T = 1:nrow(demand)
NSECost = 10000

# Get capacity factors
hours = 8760
c1, c2, c3, c4 = fill(0.85, hours), fill(1, hours), fill(1, hours), fill(1, hours)
c5 = cap_factor.Wind;
c6 = cap_factor.Solar;
c = vcat(
    permutedims(c1),
    permutedims(c2),
    permutedims(c3),
    permutedims(c4),
    permutedims(c5),
```

```

    permutedims(c6)
)

# Code from lecture, but change capacity constraint
gencap = Model(HiGHS.Optimizer)
set_silent(gencap)
# define variables
@variables(gencap, begin
    x[g in G] >= 0
    y[g in G, t in T] >= 0
    NSE[t in T] >= 0
end)
@objective(gencap, Min,
    sum(gens[G, :FixedCost] .* x) + sum(gens[G, :VarCost] .* sum(y[:, t] for t in T)) + NSEC
)
@constraint(gencap, load[t in T], sum(y[:, t]) + NSE[t] >= demand.Demand[t])
@constraint(gencap, availability[g in G, t in T], y[g, t] <= (x[g] * c[g,t]))
@constraint(gencap, emissions_constraint, sum(gens[G, :Emissions] .* sum(y[:,t] for t in T))
optimize!(gencap)

# Display results
for i in 1:6
    println("Generator Type ", i, ": ", value(x[i]), " MW")
end
println("Minimum cost: ", objective_value(gencap), " dollars")
println("Non-served energy: ", sum(value.(NSE)), " MWh")

```

```

Generator Type 1: 285.5681352306723 MW
Generator Type 2: 0.0 MW
Generator Type 3: 1509.1728698942932 MW
Generator Type 4: 703.0618976373773 MW
Generator Type 5: 2548.887923366491 MW
Generator Type 6: 2103.749396826892 MW
Minimum cost: 7.853529589434903e8 dollars
Non-served energy: 332.2907137925944 MWh

```

```

Generator Type 1 (Geothermal): 285.5681352306723 MW
Generator Type 2 (Coal): 0.0 MW
Generator Type 3 (NG CCGT): 1509.1728698942932 MW
Generator Type 4 (NG CT): 703.0618976373773 MW

```


Generator Type 5 (Wind): 2548.887923366491 MW

Generator Type 6 (Solar): 2103.749396826892 MW

Minimum cost: 7.853529589434903e8 dollars = \$785,352,958

Non-served energy: 332.2907137925944 MWh

Problem 3.3

What fraction of annual generation does each plant type produce? How does this compare to the breakdown of built capacity that you found in Problem 3.2? Do these results make sense given the demand and generator data?

```
all_plants = value.(y)
total = sum(all_plants)

for i in 1:6
    println(sum(all_plants[i,:])/total*100)
end
```

6.7785667102845935

0.0

20.300911049936303

0.7910361073940539

40.41348394350411

31.71600218886603

Generator Type 1 (Geothermal): 6.78%

Generator Type 2 (Coal): 0.0%

Generator Type 3 (NG CCGT): 20.30%

Generator Type 4 (NG CT): 0.79%

Generator Type 5 (Wind): 40.41%

Generator Type 6 (Solar): 31.72%

The fraction of annual generation is similar to the breakdown of built capacity. This makes sense because for the optimal solution, the plants with the most capacity should also tend to produce the most energy. When comparing generation, the NG CT plant has a lower than expected share compared to its relative capacity. This makes sense because NG CT has the highest variable cost, and it should only be used when the demand cannot be met by other means.

Problem 3.5

What is the shadow price of CO₂? What is the value to the utility be of allowing it to emit an additional 1000 tCO₂/yr?

```
@show shadow_price(emissions_constraint)
```

```
shadow_price(emissions_constraint) = -182.77193609185863
```

```
-182.77193609185863
```

The shadow price of CO₂ is -\$182.77. This means that if NY State were to allow one more ton of CO₂/year to be emitted, the price would go down by \$182.77.

If the utility allowed an additional 1000 tCO₂/yr, the total cost would go down by \$182,771.94 yearly.

Significant Digits

Use `round(x; digits=n)` to report values to the appropriate precision! If your number is on a different order of magnitude and you want to round to a certain number of significant digits, you can use `round(x; sigdigits=n)`.

Getting Variable Output Values

`value.(x)` will report the values of a JuMP variable `x`, but it will return a special container which holds other information about `x` that is useful for JuMP. This means that you can't use this output directly for further calculations. To just extract the values, use `value.(x).data`.

Suppressing Model Command Output

The output of specifying model components (variable or constraints) can be quite large for this problem because of the number of time periods. If you end a cell with an `@variable` or `@constraint` command, I *highly* recommend suppressing output by adding a semi-colon after the last command, or you might find that your notebook crashes.

References

List any external references consulted, including classmates.

<https://www.desmos.com/>

<https://chatgpt.com/>

<https://latexeditor.lagrida.com/>

<https://envsys.viveks.me/fall2025/slides/lecture10-1-capacity-expansion-2.html>